3341 DISCRETE MATHEMATICS

Module - 2 Previous Year Questions

PART - A [1 mark each]

1. If A has n elements, then number of subsets of A
   $= 2^n$

2. If $|A| = m$ and $|B| = n$ then $n(A \times B)$
   $= mn$

3. If $f : A \to B$ is a function from A to B. Then A is called
   = Domain

4. Give example for a finite set
   = $A = \{2, 3, 5\}$ (you can write any finite set)

5. If $(x^2 - 1, y + 3) = (3, 4)$, find values of x and y.

   =

   $x^2 - 1 = 3$ $\quad$ $y + 3 = 4$

   $x^2 = 4$ $\quad$ $y = 4 - 3$

   $x = \pm 2$ $\quad$ $y = 1$

6. Define injective function.
   = A function $f : A \to B$ is injective or one to one, if for every $b \in B$, there exist atleast one $a \in A$ such that $f(a) = b$.

PART - B [3 marks each]

1. Find $A \cup B$ and $A \cap B$ if $A = \{2, 4, 7, 8\}$ and $B = \{4, 5, 6, 7\}$

   $A \cup B = \{2, 4, 5, 6, 7, 8\}$

   $A \cap B = \{4, 7\}$

2. Write all subsets of $A = \{1, 2, 3\}$

= number of subsets $= 2^n = 2^3 = 8$

$= \{\phi\}, \{1\}, \{2\}, \{3\}, \{1,2\}, \{1,3\}, \{2,3\}, \{1,2,3\}$

3. Let $f : R \to R$ where $f(x) = 3x + 7$ for all $x \in R$.
Prove that f is one-one.

$=$ $f(x) = 3x + 7$

$f(x_1) = f(x_2)$

$3x_1 + 7 = 3x_2 + 7$

$x_1 = x_2$

4. If $f : R \to R$ and $g : R \to R$ are given by $f(x) = \sin x$ and $g(x) = 5x^2$, then find $g \circ f(x)$.

$=$ $f(x) = \sin x$ $\quad g(x) = 5x^2$

$g \circ f(x) = g(f(x))$

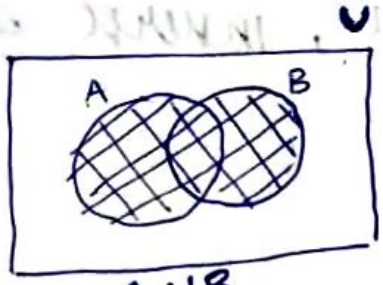
$= g(\sin x)$

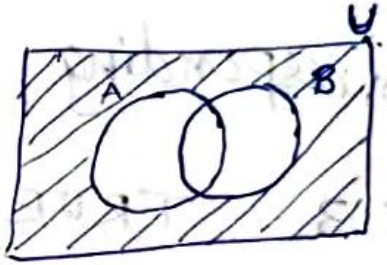
$= 5(\sin x)^2$

4. Draw appropriate Venn diagram for each

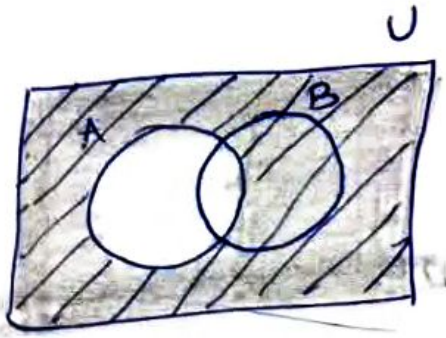
(i) $(A \cup B)'$ (ii) $A' \cap B'$ (7 marks)

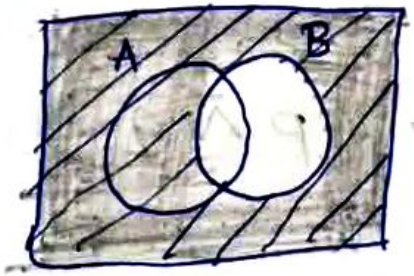
i) $(A \cup B)'$:

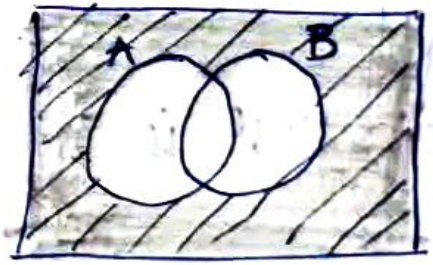




ii) $A' \cap B'$:







4- Let set $X = \{1,2,3,4\}$ and relation $R = \{(x,y) \mid x > y\}$
Draw the graph of R and also give its matrix.

Answer:

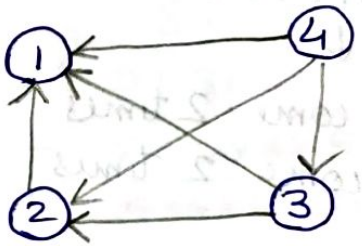
Given $X = \{1,2,3,4\}$

$X \times X$ (cross product) $= \{(1,1)(1,2)(1,3)(1,4),(2,1),(2,2),(2,3),(2,4)$
$(3,1),(3,2),(3,3),(3,4),(4,1),(4,2),(4,3),(4,4)\}$

relation $R = \{(x,y) \mid x > y\}$

So $R = \{(2,1),(3,1),(4,1),(3,2),(4,2),(4,3)\}$ (1)

Graph Representation of R



(1)

Matrix Representation of R

	1	2	3	4
1	0	0	0	0
2	1	0	0	0
3	1	1	0	0
4	1	1	1	0

(1)

3. If $A = \{1, 2, 3, 4\}$ give an example of a relation on A which is reflexive and transitive, but not symmetric.

Answer:

Given, $A = \{1, 2, 3, 4\}$

→ For a relation to be reflexive, it must contain all the self-pairs: $(1,1), (2,2), (3,3), (4,4)$ (1)

→ For a relation to be symmetric if $(a,b) \in R$ then $(b,a) \in R$
Now to make it not symmetric we can add an ordered pair in one direction: eg: add $(1,2)$ but do not add $(2,1)$

Now relation $R = \{(1,1), (2,2), (3,3), (4,4) (1,2)\}$ (1)
↘ Reflexive, not symmetric

→ A relation is transitive if $(a,b) \in R$ $(b,c) \in R$ then $(a,c) \in R$.
Here $(\overset{a}{1}, \overset{b}{2}) \in R$ and $(\overset{b}{2}, \overset{c}{2}) \in R$ also $(\overset{a}{1}, \overset{c}{2}) \in R$
and $(\overset{a}{1}, \overset{b}{1}) \in R$ and $(\overset{b}{1}, \overset{c}{2}) \in R$ also $(\overset{a}{1}, \overset{c}{2}) \in R$

Therefore the Relation $R = \{(1,1) (2,2) (3,3) (4,4) (1,2)\}$ is reflexive, transitive but not symmetric. (1)

PART - C [7 marks each]

1. Show that the function $f: N \to N$ given by $f(x) = 2x$ is one-one and onto.

$f(x) = 2x$

$f(x_1) = f(x_2)$

$2x_1 = 2x_2$

$x_1 = x_2$

So function f is one-one

Also given any real number y in R there exist $y/2$ in R such that $f(y/2) = 2 \cdot (y)$

Hence f is onto.

2. Explain different types of relations in sets with examples

i) Inverse Relation: Let R be a relation from set A to set B, then the inverse of R is the relation from B to A and is given by $R^{-1} = \{(y, x) / (x, y) \in R\}$

eg: If $R = \{(1,2), (3,2), (5,1)\}$

$R^{-1} = \{(2,1), (2,3), (1,5)\}$

ii) Universal Relation: A relation R on a set A is called Universal relation if $R = A \times A$

eg: if $A = \{1, 2, 3\}$

Then $R = \{(1,1)(2,2)(3,3)(1,2)(1,3)(2,1)(2,3)(3,1),(3,2)\}$

iii) Empty Relation: A relation R on a set A is called empty relation if $R = \phi$

eg: If $A = \{3, 4\}$ and $R = \{(x, y) / x + y > 9\}$

$\therefore R = \phi$

iv) Identity Relation: A relation R on a set A is called identity Relation if $R = \{(x, x) / x \in A\}$

eg: let $A = \{1, 2\}$

$R = \{(1, 1), (2, 2)\}$

v) Reflexive Relation: A relation R on a set A is called reflexive if $(x, x) \in R$ for all $x \in A$

eg: $A = \{1, 2\}$

$R = \{(1, 1), (1, 2) (2, 2)\}$

vi) Symmetric Relation:

if $(x, y) \in R$ then $(y, x) \in R$ for all $x, y \in A$

eg: $A = \{1, 2, 3\}$

$\{(1, 1), (2, 1), (1, 2), (2, 3), (3, 2), (1, 3) (3, 1) (2, 2) (3, 3)\}$

vii) Transitive Relation:

if $(x, y) \in R$ and $(y, z) \in R$ then $(x, z) \in R$ for all $x, y, z \in A$

eg: $A = \{1, 2\}$

$R = \{(1, 1), (1, 2) (2, 1), (2, 2)\}$

5. If A and B are two sets such that $A \cup B$ has 50 elements, A has 28 elements and B has 32 elements. How many elements does $A \cap B$ has? (7 marks)

$n(A) = 28$

$n(B) = 32$

$n(A \cup B) = 50$

$n(A \cap B) = ?$

$n(A \cap B) = n(A) + n(B) - n(A \cup B)$ [∵ inclusion-exclusion principle]

$= 28 + 32 - 50$

$= 60 - 50$

$= 10$

6. Example for a finite set (1 mark)

$A = \{1, 2, 3, 4, 5\}$

7. Among a group of students, 49 study Physics, 37 study English and 21 study Biology. If 9 of them students study Physics and English, 5 study English and Biology, 4 study Physics and Biology and 3 study Physics, English and Biology, find number of students in the group. (4 marks)

Ans: let: P → no: of students study Physics

E → no: of students study English

B → no: of students study Biology

No: of students in the group = ~~$n(P \cap$~~ $n(P \cup E \cup B)$

Given: $n(P) = 49$ $n(E) = 37$ $n(B) = 21$

$n(P \cap E) = 9$ $n(E \cap B) = 5$ $n(P \cap B) = 4$

$n(P \cap E \cap B) = 3$

From inclusion-exclusion principle:

$$n(P \cup E \cup B) = n(P) + n(E) + n(B) - n(P \cap E) - n(E \cap B) - n(P \cap B) + n(P \cap E \cap B)$$

$$= 49 + 37 + 21 - 9 - 5 - 4 + 3$$

$$= \underline{92}$$

b) Find the inverse for $f(x) = 6x + 3$. Verify your inverse by computing of the composition.

$$y = f(x) = 6x + 5$$

$$y - 5 = 6x$$

$$x = \frac{y-5}{6}$$

Inverse of f is $f^{-1}(x) = \frac{x-5}{6}$

given $f(x) = 6x + 5$

$$f^{-1}(x) = \frac{x-5}{6}$$

$$f \circ f^{-1}(x) = f(f^{-1}(x)) = f\left(\frac{x-5}{6}\right)$$

$$= 6\left(\frac{x-5}{6}\right) + 5$$

$$= x$$